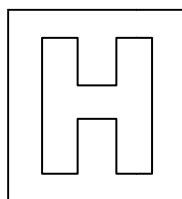


Candidate Name: _____

Class Adm No

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2014 Preliminary Examination II Pre-university 3

H2 Biology

9648/02

Paper 2

16 September 2014

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions on the question booklet.

Section B

Answer **one** question on writing paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
Total	

This question paper consists of 18 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

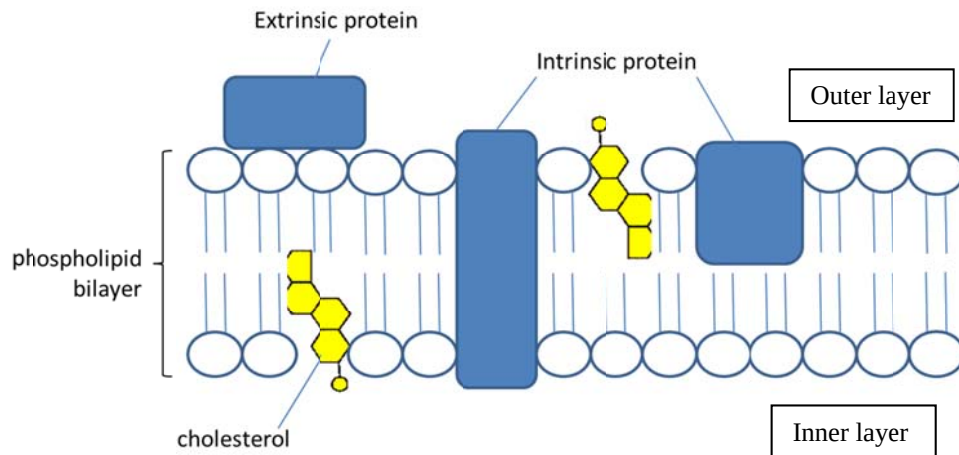
1. The fluid mosaic model of membrane structure was proposed in 1972 to describe in detail how the components of the cell membrane are organised.

(a) Some components of the cell membrane include:

- phospholipid molecules
- extrinsic protein molecules
- intrinsic protein molecules
- cholesterol molecules

- (i) In the space below, draw a labelled diagram of a cross-section of the cell surface membrane to show how the components above are organised.

Label the inner and outer membranes of the membrane.



[5]

- (ii) Suggest why 'fluid mosaic' is an appropriate term to use to describe the structure of the cell surface membrane.

- Fluid: membrane components (e.g. phospholipids and proteins) can move laterally within the plane;
- Mosaic: membrane is made up of a combination of phospholipids and proteins/glycoproteins / both layers of bilayer are asymmetrical.

[2]

The insulin receptor is embedded in the cell surface membrane of specific cell types. Insulin, secreted by the alpha cells of the Islets of Langerhans, is circulated through the blood, binding to target cells that have insulin receptors.

The oestrogen receptor is found in the cytoplasm of cells.

- (b) Explain why the insulin receptor is found on the cell surface whereas the oestrogen receptor is found in the cell cytoplasm.

- Insulin hormone is hydrophilic / large protein that cannot cross hydrophobic core of membrane;
- Oestrogen is a steroid that is hydrophobic and can cross hydrophobic core of membrane.

[2]

- (c) Explain how binding of insulin to its receptor results in lowering of blood glucose levels.

- Binding of insulin triggers dimerization and cross-phosphorylation of RTK;
- Activated receptor can bind to and phosphorylate insulin receptor substrate 1;
- IRS1 can activate PI3K, triggering phosphorylation cascade;
- Trigger one of the effects:
 - o Increase in GLUT4 receptors translocation to cell surface membrane;
 - o Increase intake of glucose into cell via facilitated diffusion
 - o Increase oxidation of glucose in respiration
 - o Increase glycogenesis and decrease glycogenolysis
 - o Increase lipid and protein synthesis;
 - o Inhibition of gluconeogenesis

[4]

(d) Suggest how amplification may occur in the insulin signalling pathway.

-
-
- One kinase can phosphorylate and activate many other proteins;
 - AVP
- ... [1]

[Total: 14]

2. Fig. 2.1 shows the assembly of the transcription initiation complex at the promoter of a proto-oncogene.

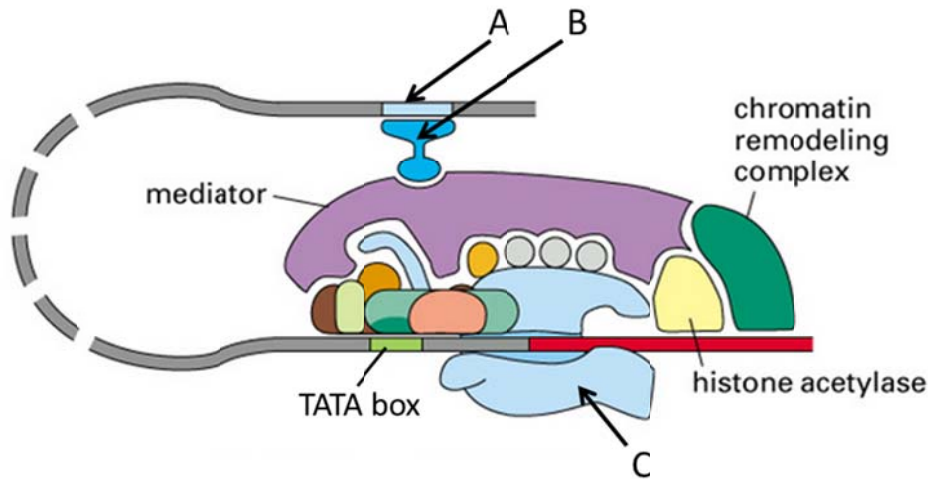


Fig. 2.1

- (a) Identify the components labelled A, B and C.

A:
 B:
 C: [1]

- A: distal control element / enhancer R silencer
- B: transcription factor / activator R repressor
- C: RNA polymerase

A mutation in the gene coding for one of the components shown in Fig. 2.1 results in the conversion of the proto-oncogene into an oncogene.

- (b) Suggest which component in Fig. 2.1 the mutation could have affected and explain how the mutation could result in cancer.

.....

 [4]

- Activator;
- Mutation resulted in hyperactive activation domain that further stabilises interaction between activator and transcription initiation complex;
- OR
- Distal control element;
- Mutation resulted in increased binding of activator to control element;
- AND
- Increases transcription rate of proto-oncogene / gain-of-function to proto-oncogene;
- Resulting in overstimulation of cell division / excessive cell division.

- (c) Suggest two differences between transcription and translation.

.....

 [2]

- Location: nucleus vs cytoplasm;
- Enzyme activity involved: RNA polymerase vs peptidyl transferase;
- Template: DNA vs mRNA;
- Product: RNA vs protein/polypeptide

Turn over

- (d) DNA polymerase is an enzyme involved in the replication of DNA. Ara-ATP is a chemical that is able to affect the enzymatic activity of DNA polymerase. Fig. 2.2 shows the structure of ara-ATP.

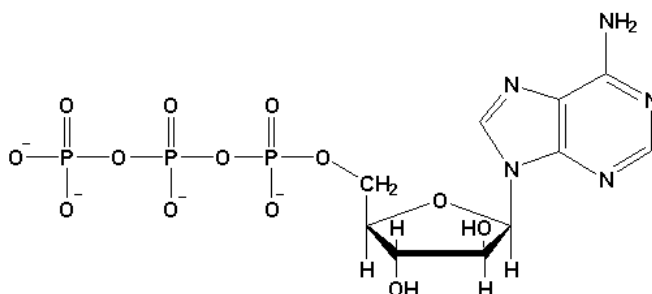


Fig. 2.2

A student decided to investigate the effect of different concentrations of ATP on the rate of DNA synthesis. The results of his experiment are shown in Fig. 2.3.

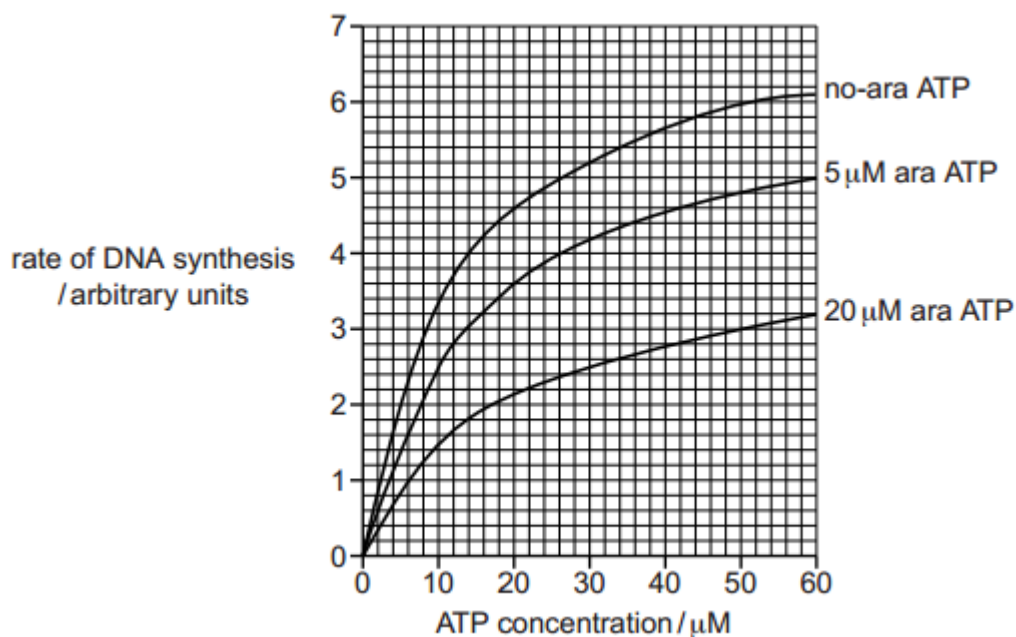


Fig. 2.3

- (i) With reference to Fig. 2.2, describe the effect of ara-ATP on DNA polymerase.

- Ara-ATP inhibits the effect of DNA polymerase as rate of DNA synthesis decreases with increasing conc. of ara ATP;
- Quote values – maximum rate of DNA synthesis is 5.9 units with no ara ATP but is 3.2 units with 20 μM of ara ATP.

[2]

(ii) Explain how ara-ATP affects DNA polymerase activity, in terms of the mode of action of enzymes.

- Ara-ATP is likely to be a non-competitive inhibitor;
- Binds to a site other than active site;
- Changes the shape of the enzyme active site to prevent substrate binding;
- Rate of enzyme activity does not reach maximum even at high levels of substrate.

[3]

(iii) State the names of two other enzymes required for DNA replication and explain their roles.

- Helicase;
- Unwind double stranded DNA from origin of replication;
- Topoisomerase;
- Make single strand nick to unwind DNA and prevent supercoiling;
- Primase;
- Synthesise short RNA primers to provide free 3'-OH end for DNA pol to begin replication
- DNA ligase;
- Catalyse formation of phosphodiester bonds to join Okazaki fragments together.

[4]

(e) Explain why there is a leading and lagging strand during DNA replication.

- DNA strands are antiparallel;
- DNA polymerase only functions in the 5' to 3' direction.

[2]

[Total: 18]

3. Woolly mammoths are extinct mammals that are related to elephants of the *Mastodon* genus. Woolly mammoths existed around 200,000 years ago during the Ice Age, where the climate changed and temperatures in the Arctic drastically decreased. These mammoths had various adaptations to reduce heat loss, such as having thick fur, small ears and small tails. During the Ice Age, the population of the woolly mammoths was significantly larger than the population of mastodons.

Woolly mammoths went extinct approximately 4000 years ago, as a result of climate change. However, the population of mastodons increased in number.

- (a) Explain how natural selection resulted in the population changes of the woolly mammoths and the mastodons.

- • Change in climate acted as selection pressure;
 - • Woolly mammoths were at a selective disadvantage and
 - mastodons were at a selective advantage as they were better
 - adapted for warm climate;
 - • Mastodons had a higher chance of reproducing and passing the
 - advantageous alleles to their offspring;
 - • Over time, allele frequencies change such that mastodons
 - increase in number while woolly mammoths decrease in
 - number.
- [3]

Scientists have used various methods to investigate the relatedness between Asian elephants and woolly mammoths.

In the first investigation, they compared the structural features of these two species, finding that both animals have similar bone structure and tusks. In the second investigation, they compared the amino acid sequence of haemoglobin from both animals.

- (b) Explain which of the two investigations is likely to provide a more accurate determination of the relatedness of these two species.

- • Comparison of amino acid sequences;
 - • No issue of analogous structures where similar selection
 - pressures in different habitats result in development of similar
 - anatomical structures;
- [2]

Scientists studied the haemoglobin of woolly mammoths and Asian elephants to determine whether or not the differences in haemoglobin were part of the mammoth's adaptation to a cold climate.

It is advantageous for Arctic mammals to have haemoglobin whose affinity for oxygen is not significantly affected by changes in temperature. This is achieved by using substances called 'red cell effectors', which can bind to haemoglobin.

Fig. 3.1 shows the effect of temperature on the affinity of oxygen of woolly mammoth and Asian elephant haemoglobin, with and without red cell effectors.

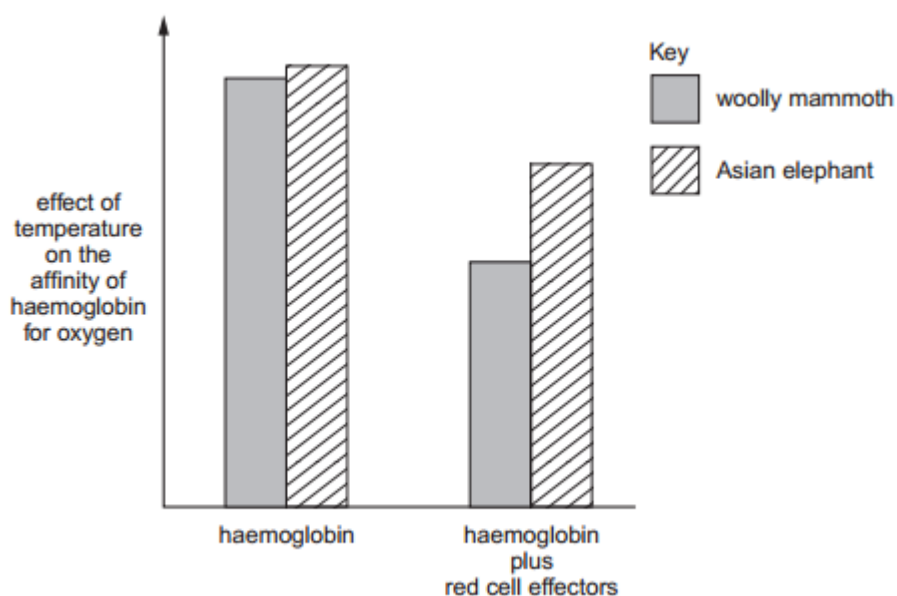


Fig. 3.1

- (c) With reference to Fig. 3.1, explain whether or not there is evidence that woolly mammoths are better adapted for a cold climate than Asian elephants.

- Yes there is evidence;
- The presence of red cell effectors reduces the effect of temperature on the affinity of haemoglobin for oxygen to a larger extent in the woolly mammoth than Asian elephant;
- Shows that natural selection selects for red cell effectors that have greater effect in cold climates.

[3]

[Total: 8]

4. In Bengal tigers, coat colour is controlled by one gene while eye colour is controlled by a separate gene. When an orange tiger with brown eyes mates with a white tiger with blue eyes, all the offspring have orange coat colour and blue eyes.

(a) Draw a genetic diagram to show the result of a cross between two tigers of the F_1 generation.

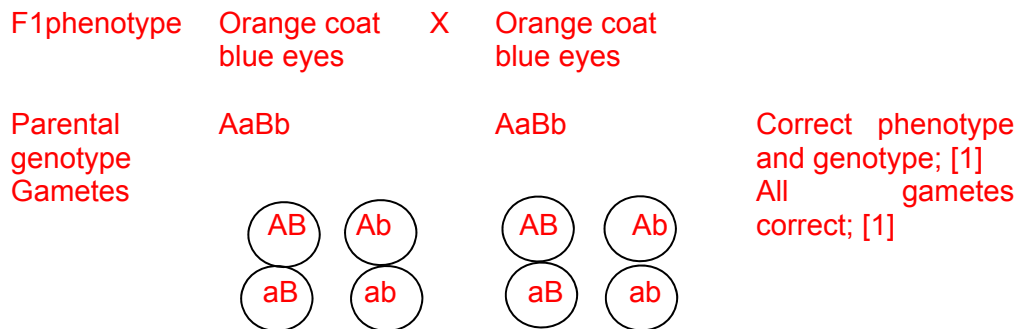
Symbols:

Let A represent the dominant allele for orange coat

a represent the recessive allele for white coat

B represents the dominant allele for blue eyes

b represent the recessive allele for brown eyes [1]



Gametes	AB	Ab	aB	ab
AB	AABB Orange coat blue eyes	AABb Orange coat blue eyes	AaBB Orange coat blue eyes	AaBb Orange coat blue eyes
Ab	AABb Orange coat blue eyes	AAbb Orange coat brown eyes	AaBb Orange coat blue eyes	Aabb Orange coat brown eyes
aB	AaBB Orange coat blue eyes	AaBb Orange coat blue eyes	aaBB White coat blue eyes	aaBb White coat blue eyes
ab	AaBb Orange coat blue eyes	Aabb Orange coat brown eyes	aaBb White coat blue eyes	aabb White coat brown eyes
Punnett square to show correct genotypes and phenotypes; [1]				
Phenotype	Orange coat blue eyes	Orange coat brown eyes	White coat blue eyes	White coat brown eyes
Offspring phenotypic ratio	9	3	3	1

- (b) Explain how it would be possible to determine whether or not an offspring from the cross in (a) with orange coat and blue eyes is heterozygous or homozygous dominant for both traits.

- Conduct a test cross with white tiger with brown eyes;
- If all offspring are orange coat and blue eyes, the parent is homozygous dominant;
- If there is a mixture of orange coat blue eyes and white coat brown eyes, then the parent is heterozygous.

[3]

- (c) A wild-life researcher conducted an investigation into the numbers of offspring for the cross conducted in (a). The numbers of offspring in the F₂ generation he observed are as follows:

Orange coat with blue eyes	16
Orange coat with brown eyes	2
White coat with blue eyes	4
White coat with brown eyes	10

- (i) Using the formula and table provided below, conduct a chi-squared test to determine whether or not the observed numbers in the F₂ generation differ significantly from the expected ratio.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

degrees of freedom	probability, <i>p</i>				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- $\chi^2 = (16 - 18)^2 / 18 + (2 - 6)^2 / 6 + (4 - 6)^2 / 6 + (10 - 2)^2 / 2$
 $= 0.222 + 2.667 + 0.666 + 32$
 $= 35.6$ (to 3sf)
- Degree of freedom = 3
- Since calculated value is greater than critical value (7.82), we can conclude with 95% confidence that observed ratio differ significantly from expected ratio.

[3]

(ii) Suggest an explanation for the results you have obtained in part (i).

-
- The genes for eye colour and coat colour are linked / found on the same chromosome.
-
-[1]

The tiger, *Panthera tigris*, is classified as an endangered species by the International Union for the Conservation of Nature and Natural Resources (IUCN).

(d) Complete Table 4.1 to show the classification of the tiger.

Table 4.1

Kingdom	Animalia
Phylum	Chordata
Class	Vertebrata
Order	Mammalia
Family	Carnivora
Genus	<i>Panthera</i>
Species	<i>Panthera tigris</i>

[3]

- (e) Fig. 4.1 shows the number of tigers in the wild between 1900 and 2010.

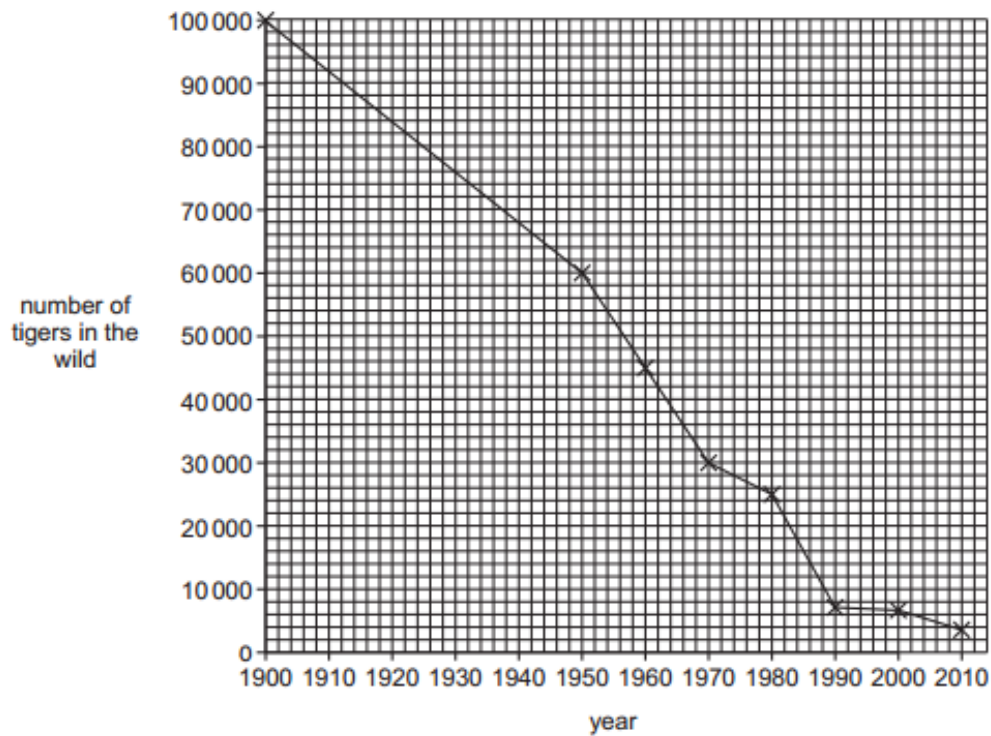


Fig. 4.1

- (i) Calculate the steepest rate of decrease observed in the tiger population between 1900 and 2010.

$$(25000 - 7000) / 10 = 1800 \text{ tiger per year}$$

..... tigers per year [1]

- (ii) Suggest a reason for the steep decrease that you have identified in part (i).

.....

 [1]

[Total: 16]

5. Table 5.1 shows the mean axon diameter and mean speed of conduction of nerve impulses for four different animals.

Table 5.1

animal		myelination of neuron	axon diameter / μm	mean speed of conduction / ms^{-1}
mammal	rabbit	myelinated	3	30
	bear	non-myelinated	4	5
amphibian	turtle	myelinated	20	121
	frog	myelinated	12	40

- (a) With reference to Table 5.1, describe and explain:

- (i) the effect of axon diameter on the speed of conduction of impulses in amphibians.

- The higher the axon diameter, the faster the rate of conduction;
- Turtle has mean speed of conduction of 121 m/s and frog has 40 m/s;
- Larger axon diameter means lower resistance to ion flow, allowing action potential to travel faster along the axon.

[3]

- (ii) the effect of myelination on the speed of conduction of impulses in mammals.

- Myelinated axons have a faster the rate of conduction than non-myelinated axons;
- Rabbit myelinated axons have mean speed of conduction of 30 m/s while bear non-myelinated axons have 5 m/s;
- Myelin sheath insulates axon and prevents ion movement in and out of axon, hence action potentials can only take place at Nodes of Ranvier.

[3]

- (b) Excessive breeding of pigeons may become an environmental issue as pigeons are able to spread disease and lower hygiene. In order to reduce the pigeon population, the practice of culling pigeons is carried out by environmental agencies, often through the use of chemicals such as Avitrol.

Avitrol is the commercial name for the chemical 4-aminopyridine. This chemical functions as an inhibitor of voltage-gated K^+ channels. Upon exposure of pigeons to Avitrol, the pigeon experiences epileptic-like convulsions.

- (i) Explain how Avitrol will affect transmission of an action potential.

- Voltage-gated K^+ channels open during repolarisation phase of action potential to allow K^+ efflux out of axon;
- Avitrol inhibits K^+ channels and prevent K^+ efflux from axon;
- Membrane potential will remain high and cannot return to resting potential, inhibiting further transmission of action potentials.

[3]

Besides inhibiting the voltage-gated K^+ channels, Avitrol can also increase the release of acetylcholine at cholinergic synapses.

- (ii) Explain how the effect of Avitrol at synapses will lead to muscle convulsions.

- Increase activation of acetylcholine receptors at post-synaptic membrane;
- Increase influx of Na^+ ions into the post-synaptic neuron;
- Increase transmission of action potentials down post-synaptic neuron that will overexcite muscle cells.

[3]

[Total: 12]

6. Fig. 6.1 shows an electron micrograph of several plant cells.



Fig. 6.1

- (a) Identify the stages of mitosis shown in cells X and Y.

X:
Y: [1]

- (b) Describe the events that occur during the stages of the cell cycle identified in cells X and Y.

..... Anaphase
.....
.....
.....
..... Max 2
..... Prophase
.....
.....
.....
..... Max 2
..... [4]

(c) Explain the process of cytokinesis in plants.

- Vesicles containing cell wall material line up at metaphase plate / equator;
- Vesicles fuse to form cell plate;
- Cell plate eventually joins up with existing parental cell wall, forming two separate cells.

[3]

(d) Colchicine is a drug that is frequently used on crops to induce formation of polyploid plants.

(i) State the meaning of the term 'polyploid'.

- Having more than the diploid set of chromosomes

[1]

(ii) Explain how colchicine may result in the formation of polyploidy plants.

- Colchicine induces non-disjunction;
- Failure of homologous chromosomes or sister chromatids to separate during anaphase of meiosis;
- Uneven distribution of chromosomes in daughter cells – cells may have more than the diploid set of chromosomes.

[3]

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

7. (a) Explain how the structure of the chloroplast contributes towards its function in carbohydrate synthesis. [6]

- Double membrane compartmentalises chloroplast into stroma and thylakoid space;
- Presence of third membrane system for lamella/grana, which increases the surface area for attachment of photosystems;
- for attachment of many electron carriers / electron transport chain and enzymes such as ATP synthase;
- These can be attached in an ordered sequence for improved efficiency;
- Photosystems have ordered arrangement of pigments & enzymes for efficient capture of light energy;
- Thylakoid membrane are impermeable to H^+ ;
- these help to maintain electrochemical gradient between the thylakoid space and stroma;
- for chemiosmosis;
- ATP synthase on lamella are involved in the production of ATP
- the stroma of chloroplast contains enzymes and other substances required for Calvin cycle

Any 6

- (b) Compare and contrast between the processes of oxidative phosphorylation and photophosphorylation. [6]

<u>Feature</u>	<u>Photophosphorylation</u>	<u>Oxidative Phosphorylation</u>
Location	Thylakoid membrane of chloroplast.	Inner membrane of the mitochondrion.
Involvement of light energy	Light energy required for splitting of water.	No light energy required.
Source of energy for ATP synthesis	Energy for ATP synthesis comes ultimately from light.	Energy for ATP synthesis comes from glucose oxidation.

Electron donors	Electron donor in non-cyclic pathway = water. AND Electron donor in cyclic pathway = PS I	Electron donors = NADH and FADH ₂ .
Final electron acceptors	Final electron acceptor in non-cyclic pathway = NADP ⁺ AND Final electron acceptor in cyclic pathway = PS I	Final electron acceptor = oxygen.
Establishing proton gradient for ATP synthesis	H ⁺ ions are pumped inwards (from stroma into the thylakoid space.)	H ⁺ ions are pumped outwards (from matrix into the intermembrane space.)

[Max 4]

Similarities

- Involve the transport of electron through an electron transport chain.
- Energy released from electron transport is used to create a proton gradient.
- The potential energy of the proton gradient is used for the synthesis of ATP from ADP and P_i.

[Max 2]

(c) Discuss how structure is related to function for starch and cellulose.

[8]

Starch

- Large and insoluble in water due to hydrophilic hydroxyl groups projecting into interior;
- No hydrogen bonds can be formed between glucose residues and water molecules;
- Can be stored in large quantities without affect water potential of cell;
- Can be prevented from diffusing out of cell;
- Individual chains can be folded into compact shape as chains are not cross-linked;
- Easily hydrolysed to form monosaccharides because of highly branched structures;
- α1,4 glycosidic bonds easily hydrolysed by enzymes like amylase;

[Total: 20]

Cellulose

- High tensile strength due to cross linking of chains by extensive hydrogen bonds;
- To form macrofibrils;
- Due to –OH groups projecting outwards in all directions;
- Large intermolecular spaces between macrofibrils to allow water and soluble materials to pass through;
- β 1,4 glycosidic bonds not easily hydrolysed

OR

8. (a) Explain how the reproductive cycle of the bacteriophage allows it to contribute towards genetic variation in bacteria.

[6]

- Transduction involves the transfer of bacterial genes from one cell to another via a bacteriophage;
- Ref. to generalised transduction involving transfer of any gene to another bacterium;
- When phage injects its genome into host cell, it hydrolyses bacterial DNA;
- During assembly stage, phage packages newly synthesised copies of its viral genome into phage capsid;
- A portion of fragmented bacterial DNA could be packaged in phage capsid by accident;
- After phage lyses the host cell and is released, new phages can infect new host cells and transfer bacterial DNA to host cells;
- Newly introduced bacterial DNA can be integrated into new host genome via homologous recombination;
- Ref. to specialised transduction involving transfer of specific bacterial genes located close to site of prophage integration;
- Ref. to lysogenic cycle and integration of phage genome into bacterial chromosome;
- Inaccurate excision of phage genome when switching from lysogenic to lytic phase.

(b) Explain how the presence of lactose and glucose can affect bacterial growth. [10]

- Ref. to lac operon and structural genes responsible for uptake and hydrolysis of lactose;
- In the absence of lactose, lac repressor is active and binds to operator;
- Repress transcription of lac operon;
- In the presence of lactose, allolactose binds to lac repressor and inactivates it;
- Lac repressor dissociates from operator and RNA polymerase can bind, transcription is no longer repressed;
- In the presence of glucose, cAMP levels are low;
- Catabolite activator protein (CAP) is inactive and does not bind to CAP binding site;
- Transcription of lac operon not activated;
- In absence of glucose, cAMP levels are high;
- cAMP binds to CAP and activates it, CAP binds CAP binding site;
- Transcription of lac operon is activated;
- Maximum rate of transcription occurs in presence of lactose and absence of glucose;
- Lactose can be hydrolysed into monosaccharides for use in respiration, to generate ATP for cellular growth.

(c) Distinguish between the reproductive cycles of the T4 phage and influenza virus. [4]

Feature	T4 phage	Influenza
Attachment / Adsorption;	Uses tail fibres bind to specific receptor sites on wall of <i>Escherichia coli</i>	Haemagglutinin recognises and bind to sialic acid receptors on host cell
Entry / Penetration;	Shaft of tail contracts, injecting phage DNA into cell and leaving empty capsid outside	Entry via endocytosis
Replication;	Utilises host cell machinery to replicate viral DNA and transcribe and translate viral proteins	Utilises host cell machinery to replicate viral RNA, viral RNA is also used to translate viral proteins
Release;	During lytic cycle, phage directs production of an enzyme that damages bacterial cell wall, allowing fluid to enter, which causes cell to swell and lyse	Each new virus buds from cell, neuraminidase on viral surface cleaves host sialic acid receptors

AVP

[Total: 20]

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